Spatial formulas

rows $(N,3)$, Z up, column action $p' = Rp + t$, cloud $(RP^\mathsf{T})^\mathsf{T} + t = PR^\mathsf{T} + t$

Vectors

$$\|v\| = \sqrt{v \cdot v}, \quad \hat{v} = \frac{v}{\|v\|}, \quad \operatorname{proj}_b a = \frac{a \cdot b}{b \cdot b}\, b$$

$$\cos\theta = \operatorname{clip}\left(\frac{a \cdot b}{\|a\|\|b\|}, -1, 1\right) \quad \text{area}_\triangle = \tfrac{1}{2}\|a \times b\|$$

$$a \times b = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} \qquad [a,b,c] = a \cdot (b \times c)$$

2D side of directed ab: $(b-a) \times (p-a) = (b_x - a_x)(p_y - a_y) - (b_y - a_y)(p_x - a_x)$. Positive is left. Right-hand offset is the negation. Zero-length v: angle and $\hat{v}$ are undefined.

Rotations and rigid motions

Right-hand rule, radians. Unit axis k, skew K with $Kv = k \times v$:

$$K = \begin{pmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{pmatrix}$$

$R = I + \sin\theta\, K + (1 - \cos\theta)\, K^2 = \cos\theta\, I + \sin\theta\, K + (1 - \cos\theta)\, kk^\mathsf{T}$

$K^2 = kk^\mathsf{T} - I$ for $\|k\| = 1$. Do not mix the two forms.

$$R_z(\theta) = \begin{pmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad c = \cos\theta,\ s = \sin\theta$$

$+90°$ about $+Z$ sends $(1,0,0) \to (0,1,0)$. $+120°$ about $(1,1,1)$ cycles $x \to y \to z$.

Valid rotation: $R^\mathsf{T}R = I$ and $\det R = +1$. $\det = -1$ is a reflection. Apply first, then second, on column vectors:

$$R = R_2 R_1, \qquad t = R_2 t_1 + t_2$$

Inverse of $p' = Rp + t$:

$$R^{-1} = R^\mathsf{T}, \qquad t_\text{inv} = -R^\mathsf{T} t$$

Quaternion $(w, x, y, z) = (\cos(\theta/2), \sin(\theta/2)\, k)$, then convert to R. Stay with matrices unless asked.

Section frame (Z up)

Horizontal unit tangent (t_x, t_y). Columns of R are forward, left, up:

$$f = (t_x, t_y, 0),\ \ell = (-t_y, t_x, 0),\ u = (0,0,1) \quad f \times \ell = u$$

Right normal, positive offset: $r = (t_y, -t_x)$. Eastbound road: south is right. $\det R = +1$.

Point, line, segment, plane

Plane through p_0 with unit n: $n \cdot (x - p_0) = 0$.

$$d_\text{signed} = (p - p_0) \cdot n \qquad d = |d_\text{signed}|$$

Three points: $n = (b-a) \times (c-a)$, then normalize. Angle between planes = angle between normals.

Line through a toward b:

$$t = \frac{(p-a) \cdot (b-a)}{\|b-a\|^2}, \quad c = a + t(b-a), \quad d = \|p - c\|$$

Segment: clip t to $[0,1]$. Unclipped t is the infinite line. If $a \approx b$, distance is $\|p - a\|$.

Skew lines $p_1 + s\, d_1$ and $p_2 + t\, d_2$:

$$d = \frac{|(p_2 - p_1) \cdot (d_1 \times d_2)|}{\|d_1 \times d_2\|}$$

Ray $o + td$, plane (p_0, n):

$$t = \frac{(p_0 - o) \cdot n}{d \cdot n}$$

Parallel if $|d \cdot n| \approx 0$. In front of the origin: $t > 0$.

Fits

Plane, SVD. Center $C = P - \bar{p}$. Normal is the last right singular vector of C (least variance), $V_t[-1]$. Flip so the largest component is positive. General plane $ax + by + cz = d$. The graph $z = ax + by + c$ cannot represent a vertical curb face. One outlier tilts a least-squares fit.

Line, PCA. Direction = top eigenvector of the covariance. A point on the line is the centroid.

RANSAC. Sample 3 points, skip a near-zero cross product, keep the largest inlier count, refit SVD, recompute the mask.

Circle (Kåsa), plan view:

$$x^2 + y^2 = 2c_x x + 2c_y y + c, \qquad r^2 = c + c_x^2 + c_y^2$$

Use `lstsq`, not $(A^\mathsf{T}A)^{-1}$. A perfect arc recovers (c, r). The centroid of a one-sided arc sits on the visible side, and $\sim$1 cm of noise on a short arc moves the fit.

Rays and triangles

Möller–Trumbore. $e_1 = v_1 - v_0$, $e_2 = v_2 - v_0$, direction need not be a unit vector:

$$h = d \times e_2, \qquad a = e_1 \cdot h, \qquad f = 1/a, \qquad s = o - v_0$$
$$u = f(s \cdot h), \qquad q = s \times e_1, \qquad v = f(d \cdot q), \qquad t = f(e_2 \cdot q)$$

Hit when $t > \varepsilon$, $u \geq 0$, $v \geq 0$, $u + v \leq 1$. $|a| \approx 0$: parallel. t is the scale in $o + td$.

Boxes

AABB, per axis: $\ell = \max(0,\ \min(a_\text{max}, b_\text{max}) - \max(a_\text{min}, b_\text{min}))$.

$$\text{IoU} = \frac{\prod \ell_i}{\text{vol}\, A + \text{vol}\, B - \prod \ell_i}$$

A shared face has volume 0.

OBB: columns of R are the box axes in world coordinates. Local of a row point is $(p - c)\, R$, i.e. $R^\mathsf{T}(p - c)$. Inside when every $|local_i| \leq h_i$.

Barycentric Z (TIN)

Query p against triangle a, b, c in XY. $v_0 = b - a$, $v_1 = c - a$, $v_2 = p - a$, $d_{ij} = v_i \cdot v_j$.

$$\begin{aligned} D &= d_{00}d_{11} - d_{01}^2 \\ v &= (d_{11}d_{20} - d_{01}d_{21})/D \\ w &= (d_{00}d_{21} - d_{01}d_{20})/D \\ u &= 1 - v - w \end{aligned}$$

Inside, boundary included, when $u, v, w \geq 0$ and $D \not\approx 0$. Then $z = uz_a + vz_b + wz_c$. First hit wins; a shared edge of a continuous mesh agrees. A breakline constrains the mesh, not the query. Drop a face with any edge over the length cap.

Kabsch and one ICP step

P and Q correspond, same order. $H = P_c^\mathsf{T} Q_c$ is 3×3.

$$H = U\Sigma V^\mathsf{T}, \quad R = VU^\mathsf{T}$$

If $\det R < 0$, flip the last row of V^T (smallest singular vector) and rebuild R.

$$t = \bar{q} - R\bar{p} \qquad Q \approx (RP^\mathsf{T})^\mathsf{T} + t$$

One ICP step: nearest neighbor source→target, then Kabsch. True matches only if the motion is small relative to point spacing.

Camera ($+Z$ forward)

$p_c = Rp_w + t$. Intrinsics K. Pixel (u, v) from $q = Kp_c$, then $u, v = q_{0,1}/q_2$. Keep $z > 0$ and $0 \leq u < W$, $0 \leq v < H$. Image shape is (H, W); u is the column. Back-project depth stored as camera z:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = z\, K^{-1} \begin{pmatrix} u \\ v \\ 1 \end{pmatrix}$$

Do not divide by $z \leq 0$.

Polylines, station, section

Station is horizontal XY chainage. Z does not advance it. Clipped t on segment i gives station $s_i + tL_i \in [0, L]$.

$$\text{offset} = (p - c) \cdot (t_y, -t_x)$$

Positive is right of increasing station. Ties take the lowest segment index. Past an end, station clamps; offset is the lateral part of $(p - \text{endpoint})$, not $\|p - c\|$.

Vertical section at an alignment origin:

$$\text{along} = (\text{xy} - o) \cdot \hat{t}, \qquad \text{offset} = (\text{xy} - o) \cdot r$$

Keep $|\text{along}| \leq$ thickness and $|\text{offset}| \leq$ half-width. Elevation is absolute Z. Interior vertex: outgoing tangent. Final vertex: incoming tangent.

Even–odd polygon test, ray $+x$: count an edge when exactly one end is strictly above the point and the crossing is to the right. Skip $dy = 0$. The boundary is unstable. Douglas–Peucker: keep endpoints; keep an interior vertex whose distance to the span exceeds ε. Segments properly cross when each pair of endpoints is strictly on opposite sides of the other line. A shared vertex is not a crossing.

Clouds and local shape

Neighbors: $\|A_{:,\text{none},:} - B_{\text{none},:,:}\|$ is (N, M). $N = M = 10^4$ is $\sim$800 MB. Chunk, or use a KD-tree. Chamfer here: $\text{mean}_A \min_B + \text{mean}_B \min_A$.

Voxel key $\lfloor p/s \rfloor$. $\lfloor -0.1 \rfloor = -1$, but a cast toward zero gives 0. Centroid = bin sum / count. `np.add.at` accumulates repeats; `sums[idx] +=` does not. Min-Z grid: same half-open bins, row 0 at minimum Y, `np.minimum.at`.

Covariance $(X^\mathsf{T}X)/N$. `eigh` returns λ ascending; reorder $\lambda_1 \geq \lambda_2 \geq \lambda_3$.

$$L = \frac{\lambda_1 - \lambda_2}{\lambda_1}, \quad P = \frac{\lambda_2 - \lambda_3}{\lambda_1}, \quad S = \frac{\lambda_3}{\lambda_1}$$

Direction = largest eigenvector, normal = smallest. Signs are arbitrary. Pole: high L, $|\text{dir} \cdot up| \approx 1$. Ground: high P, $|n \cdot up| \approx 1$. Curb face: high P, $|n \cdot up| \approx 0$. If $\lambda_1 \approx 0$, return zeros.

Before you write

State $(N, 3)$ or $(3, N)$. Clip cos before arccos. Guard every denominator. $t < 0$ is behind the ray origin. $\det R = -1$ is a reflection. SVD and eigenvector signs flip. A centimetre tolerance is not 10^{-8}.